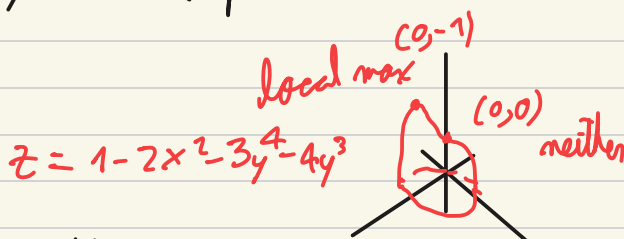


Types of critical points

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local max, local min., saddle point, neither



Theorem (2nd derivative test, expanded):

Suppose $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is twice diff at $\vec{a}$ in X and $\nabla f|_{\vec{a}} = \vec{0}$:

- 1) $f(\vec{a})$ is a local max. if $Hf|_{\vec{a}}$ is neg. def.
- 2) $f(\vec{a})$ is a local min if $Hf|_{\vec{a}}$ is pos. def.
- 3) f has a saddle point at $\vec{a}$ if $Hf|_{\vec{a}}$ has both a positive eigenvalue and a negative eigenvalue
- 4) if Hf has 0 eigenvalues and positive eigenvalues (pos. semidefinite), or 0 eigenvalues and negative eigenvalues (neg. semidefinite), or all 0 eigenvalues, then the 2nd derivative test is inconclusive

* inconclusive just means inconclusive

e.g. $f(x) = x^4$ has a local min at $x=0$
 $f'(x) = 4x^3$ $f'(0) = 0$
 $f''(x) = 12x^2$ $f''(0) = 0$



proposition (2^{nd} derivative test for 2×2): Suppose
 $f: X \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ is twice diff. and $\vec{\nabla} f|_{\vec{a}} = \vec{0}$:

1) if $\det(Hf|_{\vec{a}}) > 0$ and $\text{Tr}(Hf|_{\vec{a}}) < 0$, then
 $Hf|_{\vec{a}}$ is negative definite and $f(\vec{a})$ is a local max

2) if $\det(Hf|_{\vec{a}}) > 0$ and $\text{Tr}(Hf|_{\vec{a}}) > 0$, then
 $Hf|_{\vec{a}}$ is pos. def. and $f(\vec{a})$ is a local min

3) if $\det(Hf|_{\vec{a}}) < 0$, then f has a saddle point at $\vec{a}$

4) if $\det(Hf|_{\vec{a}}) = 0$, then 2^{nd} derivative test is inconclusive

* trace: $\text{Tr}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = a + d$ $\text{Tr}\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a + e + i$

trace is the sum of the diagonal entries

* in 2×2 matrices we also have:

1) if $\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} > 0$ and $a < 0$, $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is neg. def.

2) if $\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} > 0$ and $a > 0$, $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ pos. def.

e.g. $f(x,y) = x^3 + 2x^2 + y^3 - 12y$ find and classify crit. pts.

$$\vec{\nabla} f = (3x^2 + 4x, 3y^2 - 12) = 0 \Rightarrow x(3x+4) = 0 \begin{matrix} x=0 \\ x=-\frac{4}{3} \end{matrix} \text{ and } y = \pm 2$$

crit. pts. $(0, 2), (0, -2), (-\frac{4}{3}, 2), (-\frac{4}{3}, -2)$

$$Hf = \begin{pmatrix} 6x+4 & 0 \\ 0 & 6y \end{pmatrix}$$

$$\text{at } (0, 2): Hf|_{\vec{a}} = \begin{pmatrix} 4 & 0 \\ 0 & 12 \end{pmatrix} \quad \det(H) = 48 > 0$$

$$H_{11} = 4 > 0$$

$$\tau_1(H) = 4 + 12 = 16 > 0$$

$Hf|_{(0,2)}$ is pos. def.

$f(0,2)$ is local min.

$$\text{at } (0, -2): Hf|_{\vec{a}} = \begin{pmatrix} 4 & 0 \\ 0 & -12 \end{pmatrix} \quad \det(H) = -48 < 0$$

f has a saddle point at $(0, -2)$

$$\text{at } (-\frac{4}{3}, 2): Hf|_{\vec{a}} = \begin{pmatrix} -4 & 0 \\ 0 & 12 \end{pmatrix} \quad \det(H) = -48 < 0$$

f has a saddle point at $(-\frac{4}{3}, 2)$

$$\text{at } (-\frac{4}{3}, -2): Hf|_{\vec{a}} = \begin{pmatrix} -4 & 0 \\ 0 & -12 \end{pmatrix} \quad \det(H) = 48 > 0$$
$$\tau_1(H) = -16 < 0 \text{ as } H_{11} = -4 < 0$$

$Hf|_{\vec{a}}$ is neg. def.

$f(-\frac{4}{3}, -2)$ is local max

e.g.

8. Identify and determine the nature of all the critical points of $f(x, y) = 2xy - 2x^2 - 5y^2 + 4y - 3$

$$\vec{\nabla} f = (2y - 4x, 2x - 10y + 4) = 0$$

$$y = 2x$$

$$2x - 20x + 4 = 0$$

$$4 = 18x$$

$$y = \frac{4}{9}$$

$$x = \frac{2}{9}$$

critical point $(\frac{2}{9}, \frac{4}{9})$

$$Hf = \begin{pmatrix} -4 & 2 \\ 2 & -10 \end{pmatrix} = H$$

$$\det(H) = (-4)(-10) - (2)(2) = 40 - 4 = 36 > 0$$

$$\text{Tr}(H) = (-4) + (-10) = -14 < 0$$

$$\text{or } H_{11} = -4 < 0$$

H is negative definite $\Rightarrow f(\frac{2}{9}, \frac{4}{9})$ is a local max.

*

if n is even.

prop: In general, whenever $f: \Sigma \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is twice diff.
with n even and $\vec{\nabla} f|_{\vec{a}} = 0$, then
if $\det(Hf|_{\vec{a}}) < 0$ then f has a saddle point at $\vec{a}$.

* $\det(H) =$ product of the eigenvalues of H

* $\text{Tr}(H) =$ sum of the eigenvalues of H

Additional info. not required in this class:

a few results about eigenvalues of diagonal matrices

Addendum: • In a diagonal matrix H , the eigenvalues of H are the entries along the diagonal.

e.g. $H = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$ eigenvalues are 1, 2, 3

3 appears twice, so we say its multiplicity is 2

• In any square matrix H :

1) $\det(H) = \text{product of eigenvalues}$ (counting multiplicity)

e.g. $\det \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix} = 1 \cdot 2 \cdot 3 \cdot 3 = 12$

multiplicity 2

2) $\text{Tr}(H) = \text{sum of eigenvalues}$ (counting multiplicity)

defined as the
sum of the diagonal
entries

e.g. $\text{Tr} \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix} = 1 + 2 + 3 + 3 = 9$

multiplicity 2

- If all the eigenvalues of a symmetric matrix H are positive, then H is positive definite.
- If all the eigenvalues of a symmetric matrix H are negative, then H is negative definite.

def: A symmetric matrix is called positive semi-definite if all its eigenvalues are positive or 0.

def: A symmetric matrix is called negative semi-definite if all its eigenvalues are negative or 0.